

• Eigen values of similar matrices are same.

* If A and B are two similar matrices s.t. $A = P^{-1}BP$.
 If $\underline{v}$ is eigen vector of A corresponding to eigen value $\underline{\lambda}$, then Pv is eigen vector of B corresponding to eigen value λ .

$$Av = \underline{\lambda v} \quad A = P^{-1}BP$$

$$\Rightarrow PA = BP$$

$v \neq 0$
 $Pv \neq 0$
 as P is invertible matrix

$$B(Pv) = (BP)v = (PA)v = P(Av) = P(\lambda v) = \lambda(Pv)$$

Pv is eigen vector of BP corresponding to eigen value λ .

* How to find Eigen values?

Characteristic Polynomial :- Let A be an $n \times n$ matrix. Then $\det(xI - A)$ is called characteristic polynomial of A .

Characteristic Equation of A $\det(xI - A) = 0$.

* Eigen values of matrix A are roots of Characteristic Equation of A .

• To find Eigen values of a linear transformation T $T: V \rightarrow V$
 • First compute matrix T w.r.t any basis of V .
 • Say A is matrix of T w.r.t B .
 • Eigen values of T are same as Eigen values of A .

Example $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(x, y) = (2x, x+y)$
 $B = \{(1, 0), (0, 1)\}$, $M_B(T) = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} = A$

Characteristic Equation of A
 $\det(xI - A) = 0$

$T(1, 0) = (2, 1)$
 $T(0, 1) = (0, 1)$

$$\det\left(x \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}\right) = 0$$

$$\begin{vmatrix} x-2 & 0 \\ -1 & x-1 \end{vmatrix} = 0 \Rightarrow (x-2)(x-1) = 0$$

$$x = 2, 1$$

$\lambda = 2$

$$A \begin{bmatrix} x \\ y \end{bmatrix} = 2 \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \end{bmatrix} \Rightarrow \begin{matrix} 2x = 2x \\ x+y = 2y \\ \Rightarrow x = y \end{matrix}$$

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ eigen vector of A corresponding to
 Eigen value $\lambda = 2$.

$\lambda = 1$

$$A \begin{bmatrix} x \\ y \end{bmatrix} = 1 \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{matrix} 2x = x \\ x+y = y \\ \Rightarrow x = 0 \end{matrix}$$

$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is eigen vector of A corresponding to
 Eigen value $\lambda = 1$.

* $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(x, y) = (y, -x)$ find its eigen values
 $B = \{(1, 0), (0, 1)\}$.

$$A = M_B(T) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$\det(xI - A) = \det \begin{pmatrix} x & -1 \\ 1 & x \end{pmatrix} = x^2 + 1 = 0$$

* $T: \mathbb{C}^2 \rightarrow \mathbb{C}^2$, $T(x, y) = (y, -x)$, $x^2 + 1 = 0$
 $x = \pm i$

Eigen vectors of T , $\lambda = i$

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = i \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} y \\ -x \end{bmatrix} = i \begin{bmatrix} x \\ y \end{bmatrix}$$

$$y = ix$$

$$\begin{aligned} y &= ix \\ -x &= iy \end{aligned}$$

$$x = 1$$

$$y = i$$

$\begin{bmatrix} 1 \\ i \end{bmatrix}$ is eigen vector of T w.r.t eigen value $\lambda = i$.

$$\lambda = -i \quad \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = -i \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{aligned} y &= -ix \\ -x &= -iy \end{aligned} \quad \begin{aligned} y &= i \\ x &= i \end{aligned} \quad \begin{bmatrix} i \\ 1 \end{bmatrix}$$

$\begin{bmatrix} i \\ 1 \end{bmatrix}$ is eigen vector of T w.r.t eigen value $\lambda = -i$.

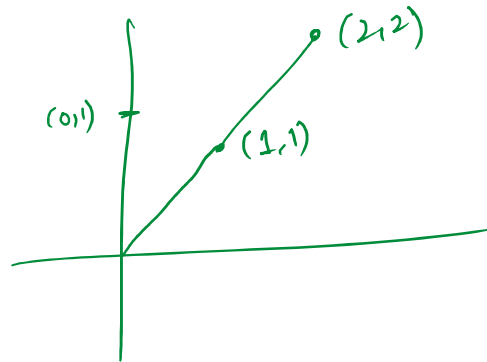
Geometric meaning of Eigen vectors.

$$T(v) = \alpha v$$

$$T(x, y) = (2x, x+y)$$

$$T(1, 1) = (2, 2)$$

$$T(0, 1) = (0, 1)$$



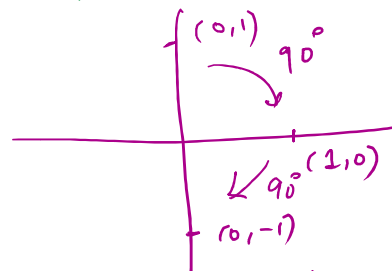
$$\bullet \quad T(x, y) = (y, -x)$$

$$T(1, 0) = (0, -1)$$

$$T(0, 1) = (1, 0)$$

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$\theta = 90^\circ$$



Rotation matrix

$$A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

n rotates vector m

$$\theta = 90^\circ$$

L
 A rotates vectors in
 clockwise direction
 by angle θ .

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(x, y, z) = (x, x+y, y-z)$$

$$B = \{ (1, 0, 0), (0, 1, 0), (0, 0, 1) \}$$

$$A = M_3(T) = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

Characteristic Equation of A

$$\det(xI - A) = \det \begin{pmatrix} x-1 & 0 & 0 \\ -1 & x-1 & 0 \\ 0 & -1 & x+1 \end{pmatrix} = 0$$

$A.M(1) = 2$
 $A.M(-1) = 1$

$\begin{vmatrix} x-1 & 0 & 0 \\ -1 & x-1 & 0 \\ 0 & -1 & x+1 \end{vmatrix} = 0$

$$(x-1)(x-1)(x+1) = 0$$

* 1 is root of $\det(xI - A) = 0$ with multiplicity 2.
 * -1 is root of $\det(xI - A) = 0$ with multiplicity 1.

$$\boxed{x = 1, 1, -1}$$

Algebraic multiplicity ^(A.M) of an Eigen value λ :- Let λ be root of characteristic polynomial $\det(xI - A) = 0$ with multiplicity k then we say, algebraic multiplicity k .

of characteristic polynomial. If the algebraic multiplicity of Eigen value λ is k , then we say, k .

find Eigen vector of $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$ corresponding to Eigen value $\lambda = 1$.

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 1 \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\begin{bmatrix} x \\ x+y \\ y-z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\begin{aligned} x &= x \\ x+y &= y \Rightarrow \underline{x=0} \\ y-z &= z \Rightarrow \underline{y=2z} \end{aligned}$$

* Eigen vector corresponding to Eigen value $\lambda = 1$ is $[0, 2, 1]$.

* Eigen vector corresponding to Eigen value -1 .

$$\begin{bmatrix} x \\ x+y \\ y-z \end{bmatrix} = - \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\begin{aligned} x &= -x \Rightarrow x=0 \\ x+y &= -y \Rightarrow y=0 \\ y-z &= -z \Rightarrow \textcircled{z} \end{aligned}$$

* Eigen vector corresponding to Eigen value $\lambda = -1$ is $[0, 0, 1]$.

(G.M)
Geometric multiplicity of an Eigen value λ
= No. of linearly independent Eigen vectors
corresponding to Eigen value λ .

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

$$\lambda = 1, -1$$

$$A.M(1) = 2$$

$$G.M(1) = 1$$

$$A.M(-1) = 1$$

$$G.M(-1) = 1$$

$$\begin{array}{l}
 \text{AM}(\lambda) = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{vmatrix} \quad \text{AM}(1) = 1 \quad \text{GM}(-1) = 1 \\
 \text{GM}(1) = 1 \quad \text{GM}(-1) = 1 \\
 \text{AM}(\lambda) \geq \text{GM}(\lambda) \quad \text{for any eigen value of } \lambda.
 \end{array}$$

Example :- $T(x, y, z) = (2x, 2y, 2z)$
 $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$

$$A = M_B(T) = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - 2 & 0 & 0 \\ 0 & \lambda - 2 & 0 \\ 0 & 0 & \lambda - 2 \end{vmatrix} = (\lambda - 2)^3 = 0$$

$\lambda = 2, 2, 2$
A.M(2) = 3

Geometric Multiplicity of 2

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 2 \begin{bmatrix} x \\ y \\ z \end{bmatrix} \Rightarrow \begin{array}{l} 2x = 2x \\ 2y = 2y \\ 2z = 2z \end{array}$$

Linearly independent vectors corresponding to Eigen value $\lambda = 2$ are

$$\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

$$G.M(2) = 3.$$

A.M(2) = G.M(2) = 3

$$G.M(2) = 3.$$

$$\underline{A.M(2) = G.M(2) = 3}$$

• Let V be an n -dimensional vector space over field F .
 Let $T: V \rightarrow V$ be a linear transformation, T is diagonalizable iff

① $\det(nI - A) = 0$, where A is matrix associated with T w.r.t some basis B of T has ' n ' roots counting multiplicity in field F .

② for all Eigen values λ of T ,
 $A.M(\lambda) = G.M(\lambda)$.